

Birla Institute of Technology & Science, Pilani

First Semester 2024-25

Mid-Semester Examination (Saturday, 05 Oct.)

Course No. : EEE/ECE F311

Course Title: Communication Systems

Max. Marks : 50

Time : 90 Min.

Name:

ID No.

Instructions: Use notation consistently. You may use standard results or formulae by stating them explicitly. Provide key intermediate steps. Highlight simplified final answers in rectangular boxes. Please avoid overwriting, as overwritten answers will not be rechecked.

Q. 1. (i) Write down the general expression for a single-tone AM signal $s(t)$ in time domain. Express $s(t)$ as the sum of the carrier term, LSB term and USB term. Assume that the single-tone AM uses the following signals and parameters: message signal $m(t) = A_m \cos(2\pi f_m t)$, unmodulated carrier signal $A_c \cos(2\pi f_c t)$, and modulation index μ . Assume that $|k_a m(t)| < 1$, where k_a denotes amplitude sensitivity. [2]

(ii) Consider a single tone AM signal $s(t) = \alpha \cos(400 \pi t) + \beta \cos(360 \pi t) + \beta \cos(440 \pi t) V$, where α and β are real positive constants. Comparing with the three-term general expression for the AM signal obtained in (i): determine carrier frequency, lower side-band (LSB) frequency, upper side-band (USB) frequency. What is the bandwidth of the AM signal $s(t)$? [2.5]

(iii) Refer to part (ii). Suppose the carrier power is 128 W and the power efficiency is 20 %. Determine the values of α , β , and μ . Express $s(t)$. What is the expression for μ as a function of α and β ? [5]

(iv) Compute the average power of the AM signal $s(t)$ in (iii). Represent the value in the dB scale. [2]

(v) The figure of merit of an AM receiver depends on the modulation index μ . Compute the figure of merit. Assume that the noise at receiver is small with respect to the signal. [1]

Q. 2. The modulating signal is a single-tone given by $m(t) = \cos(2\pi f_m t)$ and is applied to a phase modulator with phase sensitivity k_p . The phase modulated signal is of the form $A_c \cos(2\pi f_c t + \psi(t))$.

(i) Write the general expression for the single-tone phase modulated signal $s(t)$. What is the maximum phase deviation? [3]

(ii) Let $A_c = 90 V$ and the maximum phase deviation is 0.1 rad. For θ very small ($\ll 1$ rad), using the approximation $\cos(\theta) \approx 1$ and $\sin(\theta) \approx \theta$, determine the simplified expression for the phase modulated signal $s(t)$. Express $s(t)$ as the sum of the carrier term, LSB term and, USB term. [2.5]

(iii) Using the three-term simplified expression for the phase modulated signal $s(t)$ obtained in (ii), determine the spectral representation $S(f)$. List all the spectral components present in $S(f)$. [5]

(iv) Determine the complex envelope of $s(t)$. Consider the general expression obtained in (i). [2]

[P.T.O]

Q. 3. Consider an ideal band-pass filter (BPF) with mid-band frequency f_c . The amplitude response of the BPF is band-limited to $f_c - W \leq |f| \leq f_c + W$, $f_c \gg W$. Let $\tilde{H}(f)$ denote the low-pass equivalent filter, given by

$$\tilde{H}(f) = \begin{cases} \exp(-j 3.6 \pi f), & -W < f < W, \\ 0, & \text{elsewhere.} \end{cases}$$

(i) Suppose $W = 900$ Hz. Determine the impulse response $\tilde{h}(t)$ of the filter. Simplify. Is the impulse response real or complex? [2.5]

(ii) Consider the pulsed RF signal $x(t) = \{rect(t)\} \times \cos(2 \times 10^8 \pi t)$. Determine $\tilde{x}(t)$, which is the complex envelope of $x(t)$. Determine $\tilde{X}(f)$, the Fourier transform of the complex envelope. [3]

(iii) The signal $\tilde{x}(t)$ is applied to the ideal filter described by $\tilde{H}(f)$ ($\tilde{h}(t)$ in time domain). The output $\tilde{y}(t)$ is given by the following expression:

$$\tilde{y}(t) = \lambda [Si\{\mu(t - \tau_\ell)\} - Si\{\mu(t - \tau_u)\}], \text{ where } Si(v) = \int_0^v \frac{\sin y}{y} dy. \text{ Determine } \lambda, \mu, \tau_\ell \text{ and } \tau_u. [4]$$

(iv) Write down the relationship between $y(t)$ and $\tilde{y}(t)$. Obtain the expression for $y(t)$ using $\tilde{y}(t)$ (obtained after substituting the values of λ, μ, τ_ℓ and τ_u) from the previous part (iii). [3]

Q. 4. [Part-(A)] Consider a wide sense stationary (WSS) random process $Y(t)$. Its autocorrelation function is denoted by $R_Y(\tau)$ and power spectral density (PSD) by $S_Y(f)$.

Prove: $R_Y(\tau) = k \int_0^\infty S_Y(f) \cos(2\pi f\tau) df$, where k is a real positive constant.

Determine the value of k . [2.5]

[Part-(B)] The output of an oscillator is modeled as a WSS random process, which is given by $Y(t) = \cos(2\pi F t + \Psi)$, where F and Ψ are independent continuous random variables.

Let $p_F(f)$ denote the probability density function of the random variable F . The random variable $\Psi \sim U[0, 2\pi)$.

(i) Obtain a single integral (simplified) expression for the autocorrelation function of $Y(t)$ in terms of $p_F(f)$. [4]

(ii) The PSD can be expressed in the following form: $S_Y(f) = a p_F^b(f)$, where a and b are real constants. Determine a and b . Note: Use the result proved in Part-(A). [3]

(iii) Suppose $p_F(f) = \frac{1}{2} \exp(-|f|)$, $f \in R$.

Determine $S_Y(f)$ for all f , $S_Y(0)$ and $R_Y(0)$. [3]

*** END OF QUESTION PAPER ***